

Assignment 2 : RUBY ON RAILS – BLOG APPLICATION

GitHub Link:- https://github.com/Ankita-Advitot/Ruby_on_Rails/tree/feature/blog_app

5 Action View

5.1 Comment Form

`@blog.comments.build`

- Automatically sets `blog_id`

`render "blogs/show"`

- Re-renders page **without a new request**

Strong params protect from unwanted fields

```
class CommentsController < ApplicationController
  before_action :set_blog

  def create
    @comment = @blog.comments.build(comment_params)

    if @comment.save
      redirect_to @blog, notice: "Comment added successfully"
    else
      render "blogs/show"
    end
  end

  private

  def set_blog
    @blog = Blog.find(params[:blog_id])
  end

  def comment_params
    params.require(:comment).permit(:content)
  end
end
```

Ctrl+L to chat, Ctrl+K to generate

Comment Form View

Detects model state (new or persisted)

Figures out correct route:

Generates CSRF token

Generates form fields with correct names

```
<%= form_with(model: blog) do |form| %>
  <% if blog.errors.any? %>
    <div style="color: red">
      <h2><%= pluralize(blog.errors.count, "error") %> prohibited this blog from being saved:</h2>

      <ul>
        <% blog.errors.each do |error| %>
          <li><%= error.full_message %></li>
        <% end %>
      </ul>
    </div>
  <% end %>

  <div>
    <%= form.label :title, style: "display: block" %>
    <%= form.text_field :title %>
  </div>

  <div>
    <%= form.label :body, style: "display: block" %>
    <%= form.textarea :body %>
  </div>

  <div>
    <%= form.submit %>
  </div>
<% end %>
```

5.2 Create Layout and Render List of Comments

app/views/layouts/application.html.erb

```

<!DOCTYPE html>
<html>
  <head>
    <title><%= content_for(:title) || "Blog App" %></title>
    <meta name="viewport" content="width=device-width,initial-scale=1">
    <meta name="apple-mobile-web-app-capable" content="yes">
    <meta name="application-name" content="Blog App">
    <meta name="mobile-web-app-capable" content="yes">
    <%= csrf_meta_tags %>
    <%= csp_meta_tag %>

    <%= yield :head %>
    <%= tag.link rel: "manifest", href: pwa_manifest_path(format: :json) %>

    <link rel="icon" href="/icon.png" type="image/png">
    <link rel="icon" href="/icon.svg" type="image/svg+xml">
    <link rel="apple-touch-icon" href="/icon.png">

    <%= stylesheet_link_tag :app, "data-turbo-track": "reload" %>
    <%= javascript_importmap_tags %>
  </head>

  <body>
    <%= yield %>
  </body>
</html>

```

app/views/blogs/show.html.erb

```

> views > blogs > show.html.erb
1  <p style="color: green"><%= notice %></p>
2
3  <%= render @blog %>
4
5  <div>
6    <%= link_to "Edit this blog", edit_blog_path(@blog) %> |
7    <%= link_to "Back to blogs", blogs_path %>
8
9    <%= button_to "Destroy this blog", @blog, method: :delete %>
10  </div>

```

5.3 Separate Listing Page for Published & Unpublished Blogs

Detects model name → blog

Finds _blog.html.erb

Loops automatically

BlogsController

```
def published
  @blogs = Blog.where(published: true)
end

def unpublished
  @blogs = Blog.where(published: false)
end
```

Ctrl+L to chat, Ctrl+K to generate

5.4 Render Options: Partials & Locals

Locals : Avoids instance variables and Cleaner, reusable views

```
<%= render partial: "comments/comment", locals: { comment: @comment } %>
```

Ctrl+L to chat, Ctrl+K to generate

5.5 Convert ERB to HAML (Explore Only)

HAML : lease syntax , indentation based , cleaner templates

5.6 How Ruby is Embedded in Templates

Tag	Meaning
<% %>	Ruby logic
<%= %>	Output

```
<%#      Comment
%>
```

5.7 Explore form_with Options

```
4
5   form_with model: @blog, local: true
6   form_with url: blogs_path
7   form_with scope: :comment
8   form_with method: :patch
9   Ctrl+L to chat, Ctrl+K to generate
```

6 CORE & SUPPORTING COMPONENTS

6.1 SERIALIZERS

When building APIs, returning models directly leads to inconsistent JSON
A **serializer** is a **presenter** for JSON.

```
sanjana-mutkiri@sanjana-mutkiri-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Traning/Ruby And Rai
ls/blog_app$ bundle install
Fetching gem metadata from https://rubygems.org/.....
Resolving dependencies...
Fetching jsonapi-renderer 0.2.2
Fetching case_transform 0.2
Installing case_transform 0.2
Installing jsonapi-renderer 0.2.2
Fetching active_model_serializers 0.10.16
Installing active_model_serializers 0.10.16
Bundle complete! 24 Gemfile dependencies, 128 gems now installed.
Use 'bundle info [gemname]' to see where a bundled gem is installed.
1 installed gem you directly depend on is looking for funding.
  Run 'bundle fund' for details
sanjana-mutkiri@sanjana-mutkiri-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Traning/Ruby And Rai
ls/blog_app$ rails generate serializer Blog
       create  app/serializers/blog_serializer.rb
sanjana-mutkiri@sanjana-mutkiri-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Traning/Ruby And Rai
ls/blog_app$
```

6.2 SERVICE OBJECT

It is plain Ruby class encapsulates one business action

6.3 ACTIVEJOB

A **framework for background jobs** that:

- Provides a unified API
- Works with Sidekiq, Resque, DelayedJob
- Decouples job logic from queue backend

```

    create app/serializers/blog_serializer.rb
sanjana-mutkiri@sanjana-mutkiri-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Traning/Ruby And Rai
ls/blog_app$ rails generate job PublishBlog
    invoke test_unit
    create test/jobs/publish_blog_job_test.rb
    create app/jobs/publish_blog_job.rb
sanjana-mutkiri@sanjana-mutkiri-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Traning/Ruby And Rai
ls/blog_app$

```

6.4 RAKE TASKS

Some tasks:

- Don't belong to controllers
- Need CLI execution
- Are admin / maintenance jobs

```

    create app/jobs/publish_blog_job.rb
sanjana-mutkiri@sanjana-mutkiri-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Traning/Ruby And Rai
ls/blog_app$ rails generate task blogs stats
    create lib/tasks/blogs.rake
sanjana-mutkiri@sanjana-mutkiri-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Traning/Ruby And Rai
ls/blog_app$

```

6.5 BENCHMARKING

Benchmark detect slow queries and includes N+1 quires

7 TESTING & API DOCUMENTATION

Testing Setup Steps

1. Add `rspec-rails`, `factory_bot_rails`, `faker`, and `rswag` gems to the Gemfile
2. Run `bundle install` to install testing dependencies
3. Run `rails generate rspec:install` to initialize RSpec
4. Configure FactoryBot in `rails_helper.rb`
5. Create factories for Blog and Comment
6. Write model specs to test validations and associations
7. Write request specs to test blog APIs
8. Use Faker in `db/seeds.rb` for sample data
9. Run tests using `bundle exec rspec`


```
group :development, :test do
  # See https://guides.rubyonrails.org/debugging\_rails\_applications.html
  gem "debug", platforms: %i[ mri windows ], require: "debug/pry"

  # Audits gems for known security defects (use config/bundler-audit)
  gem "bundler-audit", require: false

  # Static analysis for security vulnerabilities [https://brakeman.org/]
  gem "brakeman", require: false

  # Omakase Ruby styling [https://github.com/rails/rubocop-rails]
  gem "rubocop-rails-omakase", require: false

  gem 'rspec-rails'
  gem 'factory_bot_rails'
  gem 'faker'
end

group :development do
  # Use console on exceptions pages [https://github.com/rails/web-console]
  gem "web-console"
end

group :test do
  # Use system testing [https://guides.rubyonrails.org/testing.html]
  gem "capybara"
  gem "selenium-webdriver"
  gem 'rspec'
end
```

```
hp@hp-HP-EliteBook-840-G3:~/Desktop/Ruby_on_Rails/blog_app$ rails generate rspec:install
create  .rspec
create  spec
create  spec/spec_helper.rb
create  spec/rails_helper.rb
hp@hp-HP-EliteBook-840-G3:~/Desktop/Ruby_on_Rails/blog_app$
```

This creates:

- spec/ folder
- spec/spec_helper.rb
- spec/rails_helper.rb

RSpec is now ready.

```

RSpec.configure do |config|
  config.fixture_paths = %w[
    fixtures
  ]

  # If you're not using ActiveRecord, or you'd prefer not to run
  # examples within a transaction, remove the following line or
  # instead of true.
  config.use_transactional_fixtures = true

  # You can uncomment this line to turn off ActiveRecord support
  # config.use_active_record = false

  # RSpec Rails uses metadata to mix in different behaviours to
  # for example enabling you to call `get` and `post` in request
  # specs
  #
  #   RSpec.describe UsersController, type: :request do
  #     # ...
  #   end
  #
  # The different available types are documented in the feature
  # https://rspec.info/features/8-0/rspec-rails
  #
  # You can also this infer these behaviours automatically by
  # /spec/models would pull in the same behaviour as `type: :model`
  # behaviour is considered legacy and will be removed in a future
  # version
  #
  # To enable this behaviour uncomment the line below.
  # config.infer_spec_type_from_file_location!

  # Filter lines from Rails gems in backtraces.
  config.filter_rails_from_backtrace!
  # arbitrary gems may also be filtered via:
  # config.filter_gems_from_backtrace("gem name")

  config.include FactoryBot::Syntax::Methods
end

```

This lets you write `create(:blog)` instead of `FactoryBot.create(:blog)`.

```

hp@hp-HP-EliteBook-840-G3:~/Desktop/Ruby_on_Rails/blog_app$ rails generate rswag:install
generate rswag:specs:install
rails generate rswag:specs:install
create spec/swagger_helper.rb
generate rswag:api:install
rails generate rswag:api:install
create config/initializers/rswag_api.rb
route mount Rswag::Api::Engine => '/api-docs'
generate rswag:ui:install
rails generate rswag:ui:install
/home/hp/Desktop/Ruby_on_Rails/blog_app/config/initializers/rswag_api.rb:1:in `<main>': uninitialized constant Rswag (NameError)

Rswag::Api.configure do |c|
  #####
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/engine.rb:692:in `load'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/engine.rb:692:in `block in load_config_initializer'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/activesupport-8.1.2/lib/active_support/notifications.rb:212:in `instrument'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/engine.rb:691:in `load_config_initializer'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/engine.rb:645:in `block (2 levels) in <class:Engine>'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/engine.rb:644:in `each'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/engine.rb:644:in `block in <class:Engine>'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/initializable.rb:24:in `instance_exec'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/initializable.rb:24:in `run'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/initializable.rb:103:in `block in run_initializers'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/tsort-0.2.0/lib/tsort.rb:231:in `block in tsort_each'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/tsort-0.2.0/lib/tsort.rb:353:in `block (2 levels) in each_strongly_connected_component'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/tsort-0.2.0/lib/tsort.rb:425:in `block (2 levels) in each_strongly_connected_component_from'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/tsort-0.2.0/lib/tsort.rb:434:in `each_strongly_connected_component_from'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/tsort-0.2.0/lib/tsort.rb:424:in `block in each_strongly_connected_component_from'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/3.2.0/set.rb:511:in `each_key'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/3.2.0/set.rb:511:in `each'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/initializable.rb:65:in `block in tsort_each_child'
  from /home/hp/.rbenv/versions/3.2.2/lib/ruby/gems/3.2.0/gems/railties-8.1.2/lib/rails/initializable.rb:65:in `block in tsort_each_child'

```

This creates:

- spec/swagger_helper.rb
- API documentation structure

Rswag allows API tests and documentation together.

```

FactoryBot.define do

  factory :blog do

    title { Faker::Book.title }

    content { Faker::Lorem.paragraph }

    published { false }

  end

end

```

```

10.times do
  Blog.create!(
    title: Faker::Book.title,

```

```

      content: Faker::Lorem.paragraph,
      published: true
    )
  end

  10.times do
    Blog.create!(
      title: Faker::Book.title,
      content: Faker::Lorem.paragraph,
      published: false
    )
  end
end

```

```

Note that it does not fail with a 500 error.
hp@hp-HP-EliteBook-840-G3:~/Desktop/Ruby_on_Rails/blog_app$ bundle install
Fetching gem metadata from https://rubygems.org/.....
Resolving dependencies...
Bundle complete! 29 Gemfile dependencies, 138 gems now installed.
Use `bundle info [gemname]` to see where a bundled gem is installed.
1 installed gem you directly depend on is looking for funding.
  Run `bundle fund` for details
hp@hp-HP-EliteBook-840-G3:~/Desktop/Ruby_on_Rails/blog_app$ rails db:seed
Cleaning database...
Creating published blogs...
Creating unpublished blogs...
hp@hp-HP-EliteBook-840-G3:~/Desktop/Ruby_on_Rails/blog_app$

```

Run tests

```

hp@hp-HP-EliteBook-840-G3:~/Desktop/Ruby_on_Rails/blog_app$ bundle exec rspec spec/models/blog_spec.rb
..
Finished in 0.38216 seconds (files took 1.57 seconds to load)
2 examples, 0 failures

hp@hp-HP-EliteBook-840-G3:~/Desktop/Ruby_on_Rails/blog_app$

```

8 Common Mistakes

8.1 N+1 Queries

Happens when you load objects and then repeatedly query an association in a loop.
Called N+1 because:

1. First query → load N parent objects
2. Then 1 query per object → load associated data

Fix

A tool to detect N+1 queries in development

Alerts you if your code is triggering multiple queries for associations

8.2 find vs pluck

`find` in Rails

Returns an **ActiveRecord object**
All columns are loaded by default

`pluck` in Rails

Returns an **array of raw values**,
No ActiveRecord objects are instantiated